Examiner only

7. (a) A student made some copper(II) sulfate crystals by reacting copper(II) carbonate powder with sulfuric acid using the following method.

Stage **1** Measure 50 cm^3 of sulfuric acid into a beaker.

Stage **2** Add copper(II) carbonate powder, one spatula at a time, until all the acid has reacted.

Stage **3** Filter the mixture.

Stage **4** Obtain crystals from the solution.

(i) State how you would carry out Stage **4** to get the largest possible crystals. [1]

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(ii) Crystals of copper(II) sulfate could also be made using copper(II) oxide powder instead of copper(II) carbonate powder. State and explain how the observations in Stage **2** would be different. [2]

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(iii) Complete the symbol equation for the reaction between copper(II) oxide and sulfuric acid. Copper(II) sulfate is one of the products. [2]

$CuO + H_2SO_4 \longrightarrow$ .... + ...

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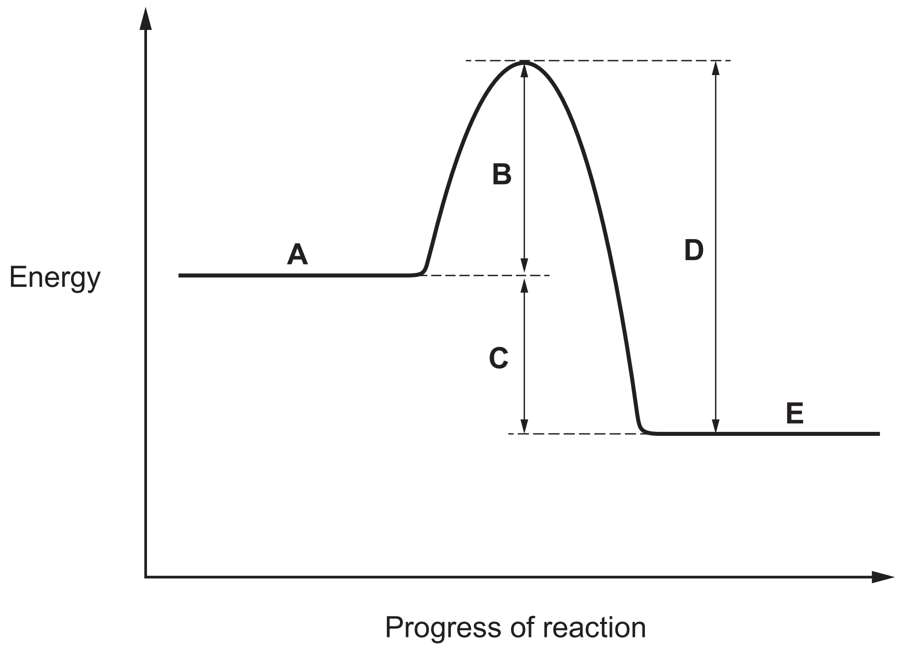
(b) The diagram shows an energy profile for a reaction.



(i) Give the **letter** that represents each of the following parts of the energy profile. [2]

Part of the energy profile	Letter
energy change for the reaction	
energy of the reactants	
activation energy of the reaction	

(ii) Give the meaning of the term activation energy. [1]

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(iii) State how the energy profile shows that this is an exothermic reaction. [1]

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